

# Lecture 2: Maxwell's Equations and EM waves

Physical principles of remote sensing  
(Ge 158)  
Fall 2026  
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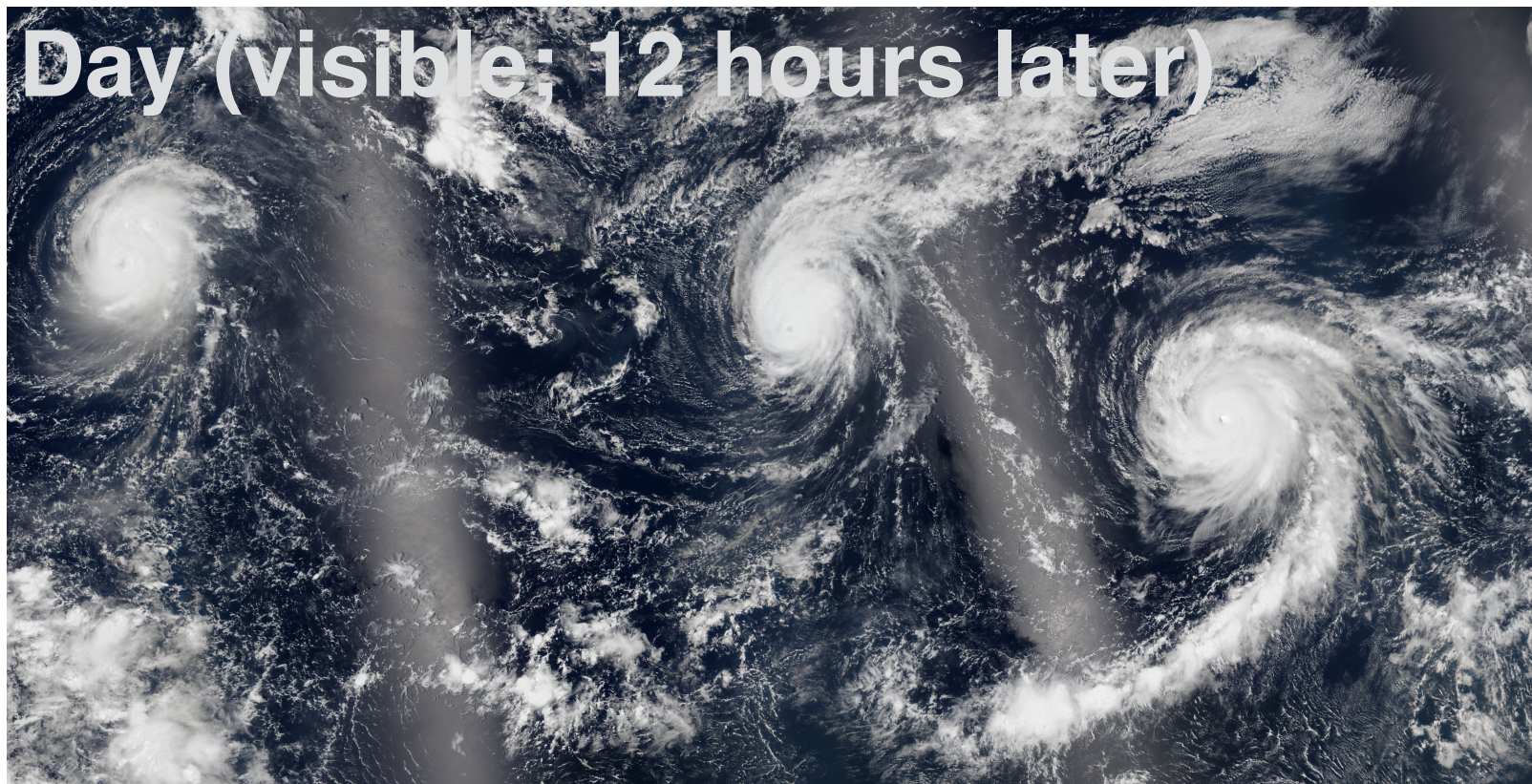
# Remote sensing image of the day

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**Night (visible to near infrared)**



**Day (visible; 12 hours later)**



3 Cat. 4 Hurricanes in the Pacific. Credit: NASA

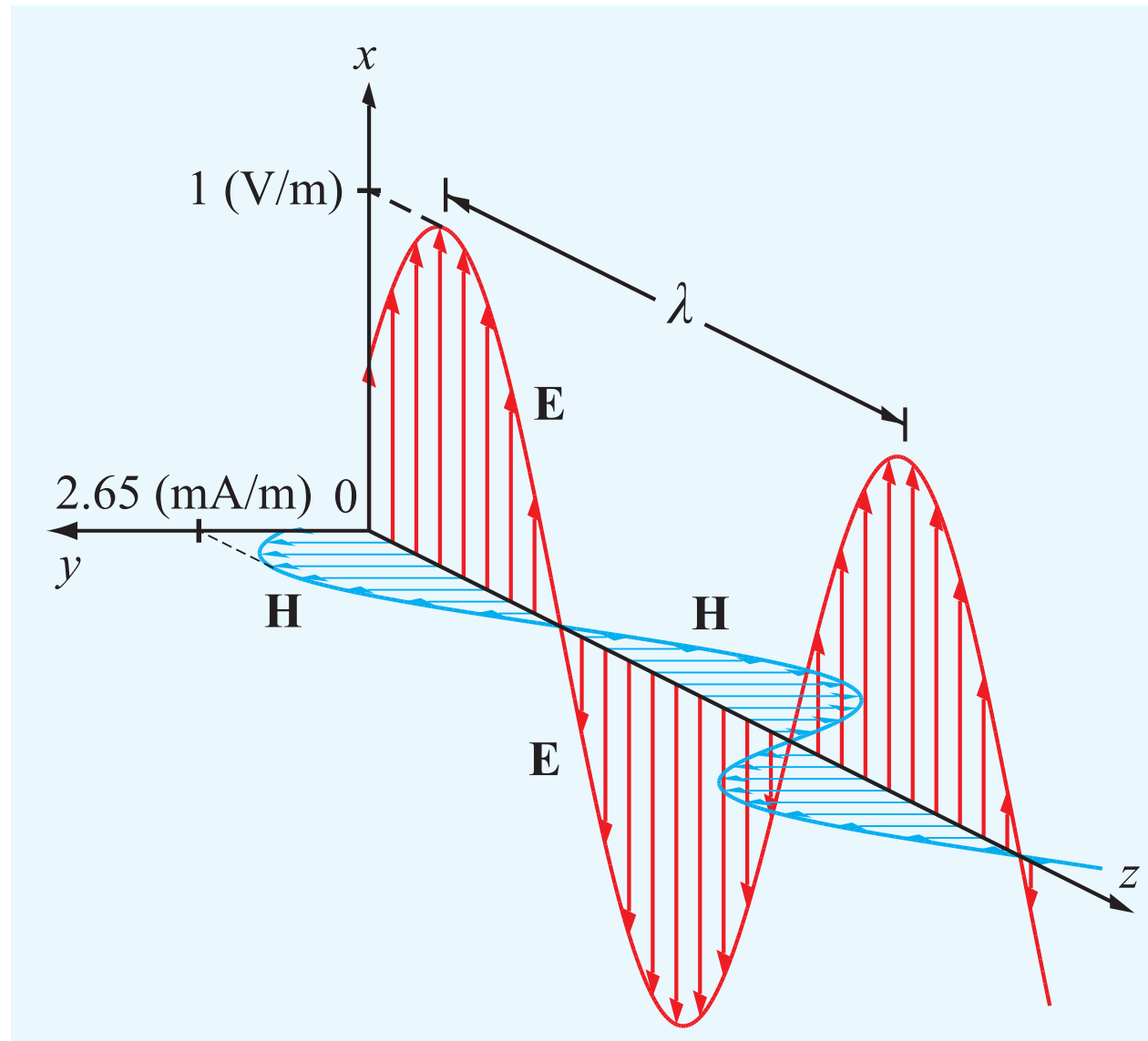
(<https://earthobservatory.nasa.gov/images/86512/trio-of-hurricanes-over-the-pacific-ocean>)

# Topics covered (vast majority covered in lecture)

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- Maxwell's Equations and EM wave equation
- Wave propagation in free space, lossless media, and lossy media
- Electromagnetic power density: Poynting vector

# EM waves in lossless medium



- Electric field **E** in red, magnetic field **H** in blue
- Note that **E** field and **H** field are in phase (see zero-crossings)
- Values (e.g., amplitude and wavelength) will change for different materials, but some properties will remain:
  - **E** and **H** are always perpendicular and always propagate along the vector **E** × **H**
  - **E** and **H** are always in phase in lossless ( $\sigma=0$ ) media; never in phase for lossy media

# Useful table

|                                                                                                                                                                                                                 | Any Medium                                                                                                                                     | Lossless Medium<br>( $\sigma = 0$ )       | Low-Loss Medium<br>( $\epsilon''/\epsilon' \ll 1$ ) | Good Conductor<br>( $\epsilon''/\epsilon' \gg 1$ ) | Units        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|-----------------------------------------------------|----------------------------------------------------|--------------|
| $\alpha =$                                                                                                                                                                                                      | $\omega \left[ \frac{\mu_0 \epsilon' \epsilon_0}{2} \left[ \sqrt{1 + \left( \frac{\epsilon''}{\epsilon'} \right)^2} - 1 \right] \right]^{1/2}$ | 0                                         | $\frac{\pi \epsilon''}{\lambda_0 \sqrt{\epsilon'}}$ | $\frac{\pi \sqrt{2 \epsilon''}}{\lambda_0}$        | (Np/m)       |
| $\beta =$                                                                                                                                                                                                       | $\omega \left[ \frac{\mu_0 \epsilon' \epsilon_0}{2} \left[ \sqrt{1 + \left( \frac{\epsilon''}{\epsilon'} \right)^2} + 1 \right] \right]^{1/2}$ | $\frac{2\pi \sqrt{\epsilon'}}{\lambda_0}$ | $\frac{2\pi \sqrt{\epsilon'}}{\lambda_0}$           | $\frac{\pi \sqrt{2 \epsilon''}}{\lambda_0}$        | (rad/m)      |
| $\eta_c =$                                                                                                                                                                                                      | $\sqrt{\frac{\mu_0}{\epsilon' \epsilon_0}} \left( 1 - j \frac{\epsilon''}{\epsilon'} \right)^{-1/2}$                                           | $\frac{\eta_0}{\sqrt{\epsilon'}}$         | $\frac{\eta_0}{\sqrt{\epsilon'}}$                   | $\frac{(1 + j) \eta_0}{\sqrt{2 \epsilon''}}$       | ( $\Omega$ ) |
| $u_p =$                                                                                                                                                                                                         | $\omega / \beta$                                                                                                                               | $c / \sqrt{\epsilon'}$                    | $c / \sqrt{\epsilon'}$                              | $c \sqrt{2 / \epsilon''}$                          | (m/s)        |
| $\lambda =$                                                                                                                                                                                                     | $2\pi / \beta = u_p / f$                                                                                                                       | $u_p / f$                                 | $u_p / f$                                           | $u_p / f$                                          | (m)          |
| Notes: In practice, a material is considered a low-loss medium if $\epsilon''/\epsilon' < 0.01$ and a good conducting medium if $\epsilon''/\epsilon' > 100$ ; $c = 3 \times 10^8$ m/s; $\eta_0 = 377 \Omega$ . |                                                                                                                                                |                                           |                                                     |                                                    |              |

From Ulaby and Long, 2014

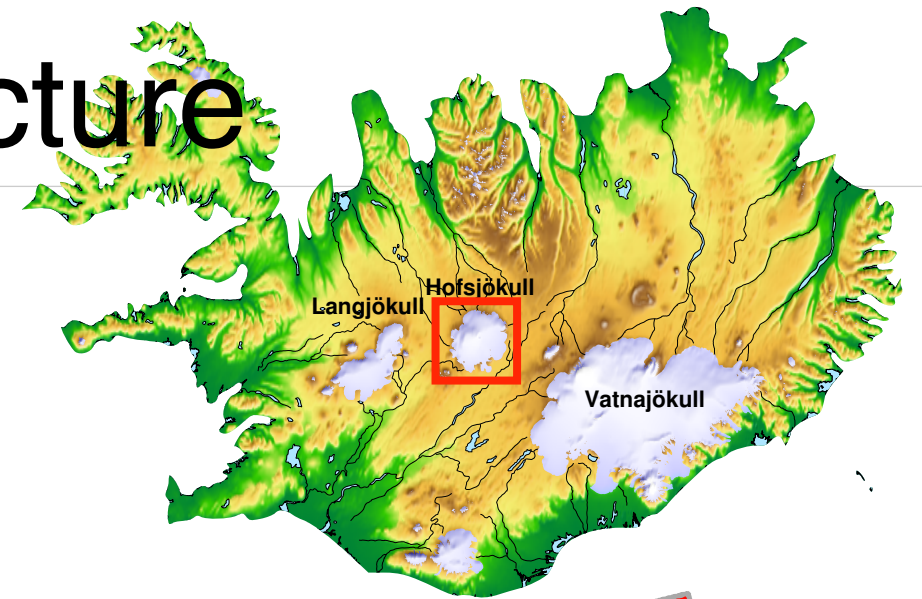
# Summary (most material covered in lecture)

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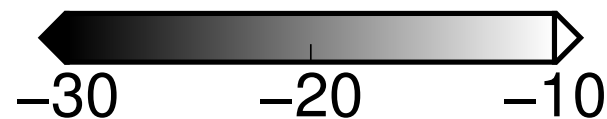
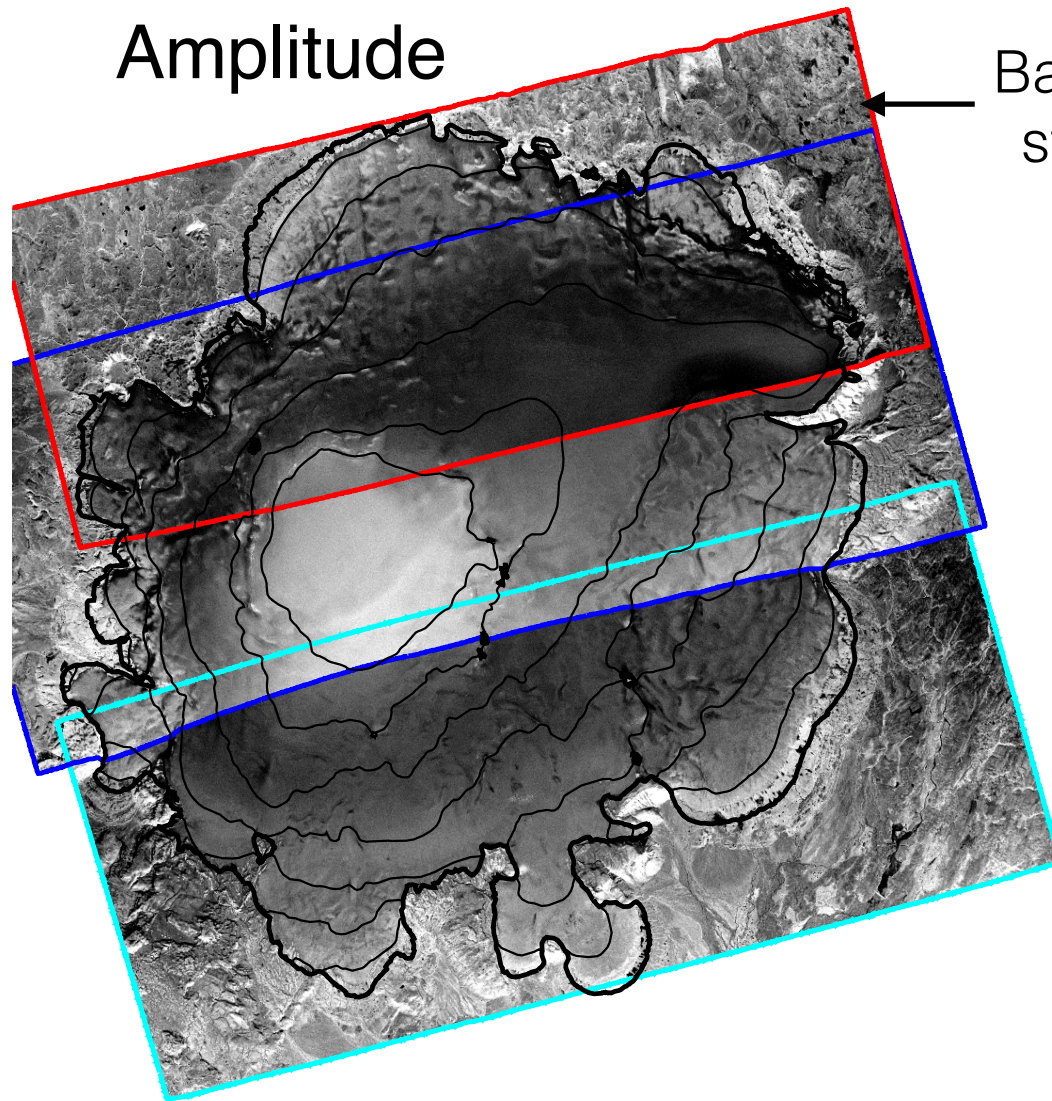
- Maxwell's Eqs.
  - Gauss's Law: in RS,  $\text{div } \mathbf{E} = 0$  because materials have neutral charge ( $\rho_v = 0$ ); "curling"  $\mathbf{E}$  field is admissible
  - Gauss's Law for magnetism:  $\text{div } \mathbf{H} = 0 \rightarrow$  there is no (known) magnetic monopole;  $\mathbf{H}$  fields tend to form closed loops)
  - Faraday's Law: temporal variations in  $\mathbf{H}$  field induce a curling  $\mathbf{E}$  field (and vice versa)
  - Ampère's Law: temporal variations in  $\mathbf{E}$  field and electrical current generate curling  $\mathbf{H}$  field (and vice versa)
- Wave Eq. and propagation of EM waves
  - Universal:  $\mathbf{E} \perp \mathbf{H}$ ;  $\mathbf{E}$  and  $\mathbf{H}$  related through intrinsic impedance
  - Free space: all EM waves travel at a constant speed =  $c$ ;  $\mathbf{E}$  in phase with  $\mathbf{H}$
  - Lossless (and low-loss) media:  $\mathbf{E}$  in phase with  $\mathbf{H}$ ; phase velocity function of permittivity  $\rightarrow$  matter is dispersive and  $v_p < c$  (except for anomalous dispersion, not discussed in this course)
  - Lossy media: Attenuation (skin and penetration depths) caused by conduction; conduction decreases phase velocity;  $\mathbf{E}$  and  $\mathbf{H}$  out of phase

# RS applications of today's lecture

Repeat-pass interferometric synthetic aperture radar (InSAR)

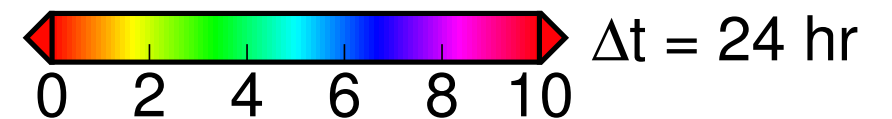
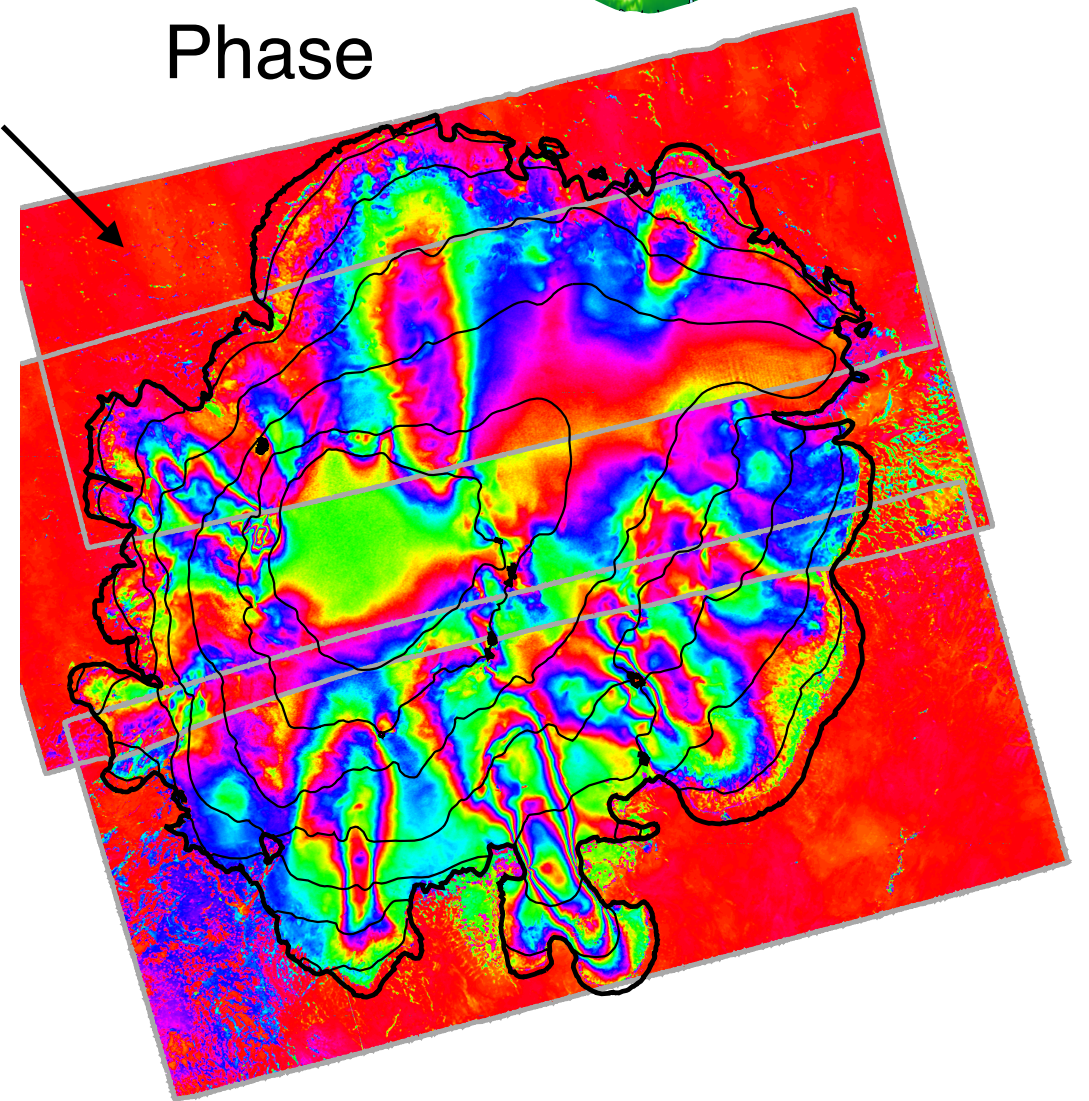


Amplitude



$\sigma_{HH}^0$  (dB)

Phase



LOS Disp. (cm)